



11x - Notes and Theoretical Background

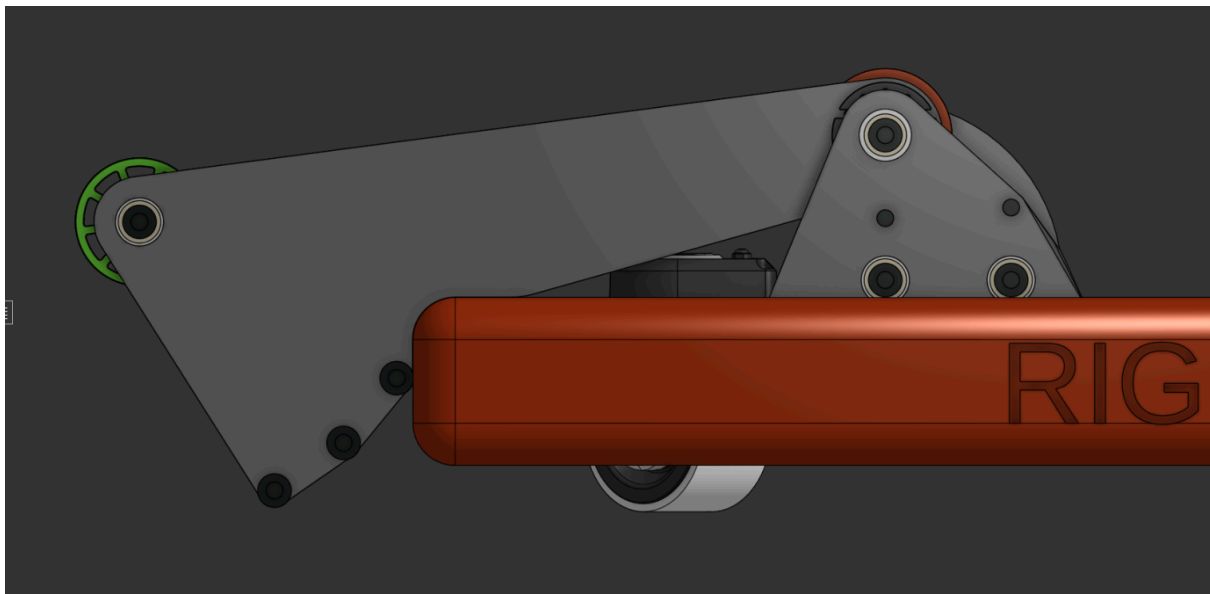
Theoretical Background:

Before we get to the slapdown intakes we first need to understand what is an intake:

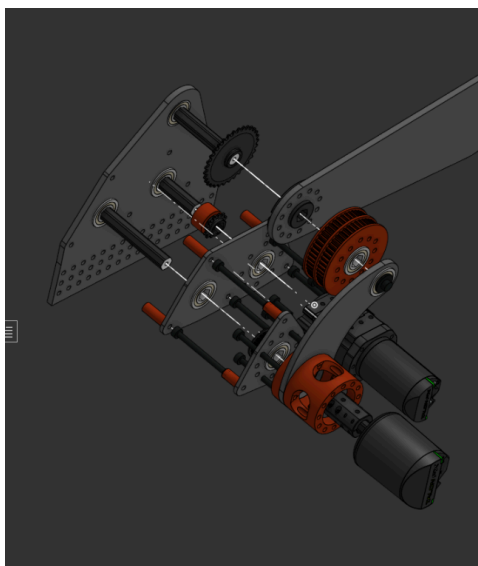
An intake is a main system in almost every year's robots. Its purpose is to acquire game pieces from outside the robot and get them into the robot perimeter and to the rest of the systems. Basically, an intake is the first thing that meets the game piece during the match.

We'll define an intake as: "The system responsible for manipulating a game piece into the robot perimeter". By this definition you'll see that some systems will fulfil this definition while not being regarded as an intake and vice versa.

A "slapdown" intake is a general name for a small family of intake designs that revolves around, well, revolving around something - in this case a rotating axis. The basic concept of a slapdown intake is constructed of two assemblies - the "static assembly" and the "rotating assembly". While the static assembly stays fully inside the robot perimeter at all times, even when the intake is deployed, the rotating assembly starts the game inside the robot perimeter and can rotate ("slap") all the way to outside the robot perimeter to deploy the intake.

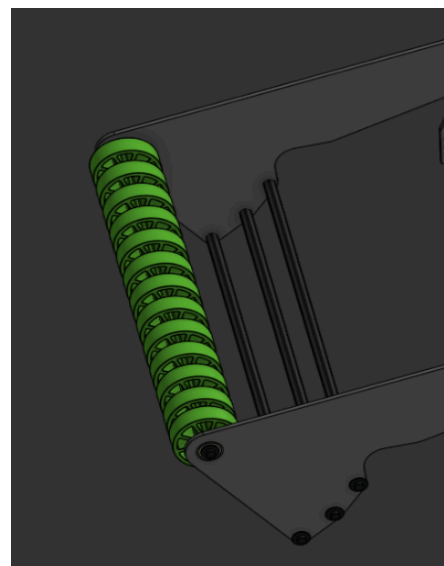


Intake deployed (mid match)



Static
assembly

Rotating
assembly

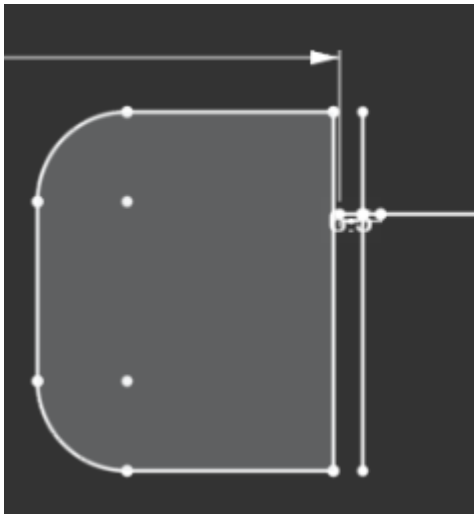




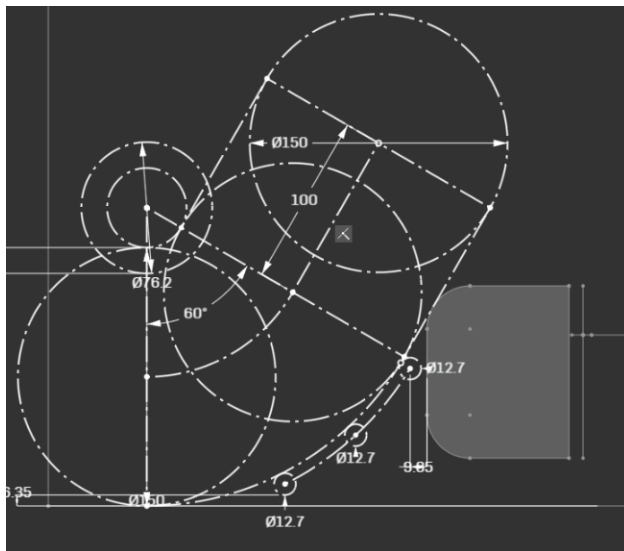
The slapdown intake family consists of two designs - Over The Bumper and Cut Bumper. In both designs the working principle is the same but the path the game piece goes through is different.

Over The Bumper:

As the name suggests, in an Over The Bumper slapdown intake the game piece starts at the floor and goes **above the bumper and into the robot perimeter**. When creating the master sketch for an OTBI¹, it is important to **consider the bumper dimensions as a constraint** or an "impenetrable wall" that blocks the straight path of the game piece. In an OTBI the path the game piece takes is most of the time defined by a curve rather than a straight line.



Bumper dimensions defined using the Use feature



2026 intake geometry - the fuel doesn't pass through the bumper

1 - Over The Bumper Intake

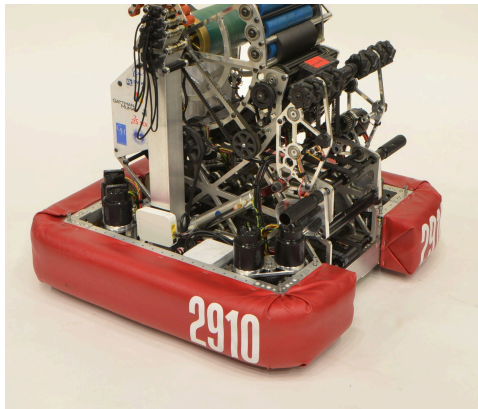


Cut Bumper Intake:

A cut bumper intake is possible only in years when the gap between the bumpers is larger than the game piece, that is because a cut bumper intake works by **passing the game piece through an opening in the bumper rather than above it**. In this file we will talk both on **Revolving** Cut Bumper Intake and **Static** Cut Bumper Intake.

Revolving:

The path the game piece takes in a RCBI² is the only major difference between it and an OTBI, as **they are both a slapdown intake by definition** (they both have static and rotating assemblies), therefore the difference in designing a RCBI is only in the path. While the intake geometry stays the same, the bumper constraint is now gone and is replaced by the inside of the robot. This means that while the bumper itself is not gone, it is no longer obstructing the path.

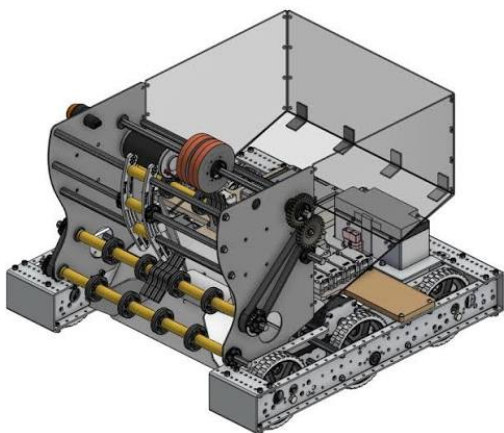


An example of a revolving cut bumper intake - 2910 in 2021.

Notice that there is still a slapdown intake above the cut part.

Static:

A SCBI³ has no revolve axis and is all part of the same static assembly, as part of that the entirety of the intake stays inside the robot perimeter at all times. Only being one static assembly means that the SCBI is not a “slapdown” intake. The geometry design for a SCBI includes only the path of the game piece and the desired compression.



An SCBI - KitBot 2026

2 - Revolving Cut Bumper Intake

3 - Static Cut Bumper Intake

